

Business Requirements Document

B2B Wholesale Marketplace & Procurement Platform — Lebanon

TradeBay is a marketplace where a buyer company asks for goods, supplier companies quote prices, one quote becomes a purchase order that freezes those prices forever, the buyer pays the platform, the platform keeps a commission and pays the rest to the supplier — and every money movement is written down as an unchangeable line in a ledger instead of being stored as an editable balance.

VERSION

2.0 — ERD aligned

DATE

September 2026

MARKET

Lebanon, domestic B2B

STATUS

Approved implementation baseline

DATA MODEL

50 collections • ERD.md v1

00 Document Control

Item	Value
Document	TradeBay Business Requirements Document
Version	2.0 (supersedes 1.1 and 1.2)
Product type	B2B marketplace + procurement + marketplace finance platform
Primary market	Lebanon only; domestic business buyers and suppliers
Authoritative data model	ERD.md — 39 commercial core + 11 extension = 50 collections
Companion documents	ERD.md · ERD.html · docs/architecture.md · docs/domain-map.md

Requirement code stability

ERD.md cites requirement codes directly in its design rules — BR-14 , FIN-AR-04 , FIN-AR-07 , FIN-AR-08 , FIN-LED-01 , FIN-LED-02 , FIN-LED-04 . Those codes keep their meaning in this version. **Do not renumber them.**

What changed in version 2.0

Change	Reason
Added the full end-to-end narrative (§06) and the exception catalogue (§07)	The previous versions described features but never walked one transaction from signup to supplier payout, and never enumerated what happens when each step fails.
Added the two ledgers as explicit requirements	v1.2 described receivables and settlement but omitted <code>financial_transactions</code> and <code>platform_transactions</code> . They are the basis of <code>FIN-LED-01..03</code> and <code>BR-14</code> .
Separated supplier <i>payable</i> from supplier <i>payout</i>	“We owe \$950” and “we sent \$950” are different facts for days. A failed payout must leave the payable open.
Added Communication as a first-class domain	General-purpose company-to-company messaging, modelled in ERD §14. v1.2 had notifications only.
Added Negotiation as a first-class domain	Counter-offer chains are distinct from quotation versioning.
Added Business Planner as a third AI capability	Implemented as plans, plan items and price estimates. v1.2 stated “exactly two AI features”.
Corrected every state lifecycle (§12)	They now match the implemented enums exactly — v1.2 listed an order state (<code>READY</code>) that does not exist and omitted the RFQ lifecycle entirely.
Withdrew loyalty, expenses, chart of accounts, favorites, supplier bills, receipts, notification templates	Carried over from v1.1 but explicitly dropped by ERD §12. Keeping them made the BRD promise things the system deliberately does not build.

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01 Executive Summary

TradeBay is a business-to-business digital marketplace for Lebanese wholesalers, manufacturers, distributors and business buyers. It combines supplier discovery, catalog and wholesale pricing, RFQs, quotations, negotiation, purchase orders, fulfilment, marketplace finance, trust and AI-assisted sourcing in one platform.

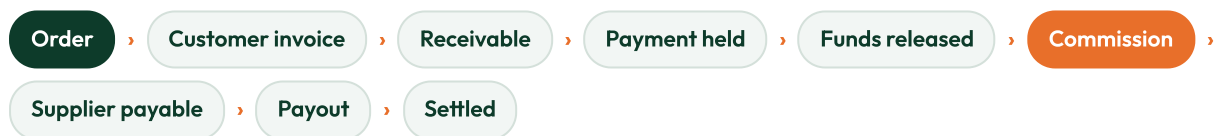
Unlike a consumer store, TradeBay is designed around **procurement**. B2B purchases are rarely “add to cart”: buyers compare suppliers, request quotes, negotiate quantities and commercial terms, agree payment conditions, and then expect accurate invoices, payments, supplier settlement and financial visibility.

Money is therefore a first-class domain rather than a reporting afterthought. Orders generate invoices and receivables; supplier obligations are tracked as payables; TradeBay commission becomes platform revenue; settlement clears what is owed; and every material movement is auditable through two append-only ledgers.

The value chain



The money chain



The governing principle. TradeBay must be built as a real business system, not a set of CRUD screens. Historical commercial and financial facts are preserved, authorization is enforced server-side on every request, financial records are traceable, and operational events drive the financial records automatically.

02 Business Problem & Opportunity

- Lebanese wholesale suppliers are fragmented across websites, social media, phone calls, messaging apps, personal networks and offline catalogs.
- Business buyers have no standardized way to discover and compare suppliers by price, MOQ, stock, lead time, verification, reputation and fulfilment performance.
- Wholesale purchasing needs RFQs, quotations, negotiation, payment terms, invoices and supplier coordination — not a fixed-price retail checkout.
- Suppliers need a digital channel for catalog exposure, RFQ opportunities, repeat orders and performance visibility.

- Platform operators need reliable financial visibility: GMV, commission, receivables, payables, settlements and refunds.
- Trust is impossible without supplier verification, transaction history, verified reviews, dispute handling and auditable financial records.

03 Objectives & Success Metrics

Objective	Success measure
Digitize wholesale discovery	Verified suppliers, active products, buyer searches, qualified RFQs.
Improve procurement efficiency	RFQ response rate, quote turnaround, acceptance rate, time from RFQ to order.
Increase transaction volume	GMV, completed orders, average order value, repeat purchase rate.
Build a sustainable platform business	Recognized commission revenue, settlement accuracy.
Strengthen trust	Verified supplier ratio, review coverage, dispute rate, cancellation rate, on-time fulfilment.
Maintain financial control	Invoice/payment reconciliation, AR and AP aging, complete audit trails, zero unexplained ledger variance.

04 Scope

4.1 In scope

Domain	Capabilities
Identity & authorization	Registration · business accounts · memberships · invitations · roles · permissions · sessions · supplier verification · audit logs
Catalog & product	Categories · products · SKU / MOQ / unit · search and filter · images · wholesale price tiers · current inventory · append-only stock ledger
Procurement & fulfilment	RFQs · RFQ items · supplier invitations · quotations · quotation items · comparison · award · purchase orders · shipments · tracking · delivery evidence
Customer finance (AR)	Invoices with embedded lines · payments · allocation across invoices · credit notes · refunds · append-only AR ledger
Platform money	Commission records · supplier payables · supplier payouts · settlement batches · append-only routing ledger
Trust & inbox	Reviews · moderation · disputes · evidence · notifications

Domain	Capabilities
Communication	Conversations · participants · messages · system timeline events
Negotiation	Structured offers · counter-offer chains · commercial versioning
AI	Sourcing Assistant · Personalized Product Recommendations · Business Planner
Settings	Platform settings · VAT settings · business letterhead

4.2 Boundary

TradeBay is a Lebanon-only domestic B2B marketplace. It is **not** an ERP, **not** a freight carrier or logistics provider, **not** a bank, and **not** a statutory accounting system. Consumer retail is not the business model. Supplier operational data is marketplace data that suppliers must keep current.

The complete exclusion list, with reasons, is **§14**.

05 Actors & Authorization Model

5.1 Actors

Actor	Responsibilities
Buyer	Discover suppliers, create RFQs, compare and negotiate quotations, order, pay, track delivery, review.
Supplier	Maintain profile and catalog, respond to RFQs, quote, negotiate, fulfil, receive settlement.
Platform staff	Verify suppliers, moderate the marketplace, resolve disputes, approve settlements, configure the platform.
System	Emit domain events, calculate totals, reserve inventory, post ledger rows, notify users. No human login.

5.2 The authorization chain

A person logs in. A company trades. A **membership** connects them. This separation is the foundation the rest of the system leans on.



Why not put a role on the user? Because one person may be Sales Manager at one company and a read-only Viewer at another. A role on the user would force two accounts and two passwords. A role on the *membership* gives one account and two contexts.

5.3 Seeded system roles

Copied onto every new business and undeletable: **Business Admin, Sales Manager, Sales Representative, Finance, Viewer**. Platform staff belong to the business account of type `platform` with **Platform Admin** or **Platform Operator**. Each company's "Sales Manager" is its own record — companies may add custom roles but cannot delete the system ones.

5.4 Principles

- **Deny by default.** Permissions are atomic `resource.action` codes, never role-name comparisons.
- **Resolved on every request**, never cached onto the user or copied onto an invitation.
- **Company scoping.** A user reaches only records belonging to the active business, unless the role carries a platform-wide permission such as `suppliers.verify`.
- **Selling requires verification.** Catalog, inventory and quotation writes need the permission code *and* a verified supplier profile.
- **No privilege escalation.** A user may only grant a permission set that is a subset of their own.
- **Financial history is append-only.** Corrections are credit notes, refunds and reversals.

06 The TradeBay Story — End to End

This section is the reference narrative for the whole product. Every requirement in §08 and §09 exists to make one of these steps work. The cast: **Ahmad**, a buyer at ABC Trading; **Lara**, a supplier at Cedar Plastics; and a **TradeBay Platform Operator**.

ACT 1 The cast — people, companies, and the link between them

Three kinds of thing exist before anything happens. **People** have an email and a password; a person logs in and never buys or sells. **Companies** buy and sell and never log in; a company is a buyer, a supplier, or the platform itself. **Memberships** are the rows that say “Ahmad works for ABC Trading as a Sales Manager.”

ACT 2 Ahmad signs up and ABC Trading is born

Ahmad registers and his account is `pending`. He verifies his email and becomes `active`. He cannot perform a commercial action until both are true. He creates ABC Trading, and three things happen together: the company is created, the five system roles are copied onto it, and Ahmad gets a membership carrying the Business Admin role.

When he logs in, a session records which company he is wearing. Switching companies changes that field, and every permission check changes with it.

Hiring an employee

Ahmad invites Sara and **picks a role**, never individual permissions. Sara accepts; if she already has an account it is matched by email, otherwise one is created. Either way a membership is created with the invited role. Because the invitation stored only a role, editing what “Sales Representative” may do later changes Sara's powers automatically.

Someone tries to promote themselves

Sara has `users.invite` but not `roles.manage`, and tries to invite a friend as Business Admin. The subset rule blocks it: you can never hand out a power you do not hold.

Platform staff

TradeBay's own employees are not special users with a magic flag. They are members of the `platform` company holding roles that carry platform-wide permissions. Platform power comes from the same machinery as everything else. Every sensitive action along the way writes an audit record.

ACT 3 Cedar Plastics becomes a seller

Lara creates Cedar Plastics the same way. It can **buy** immediately but cannot **sell**. To sell, it needs a supplier profile — exactly one per company — with trade licence documents and declared service areas. A Platform Operator reviews and verifies it.

From that moment a second gate applies to every selling action: the role must carry the permission **and** the company must be verified. An unverified company with a perfect Business Admin role still cannot list a product.

ACT 4 Cedar Plastics builds its catalog

Lara lists 5-litre plastic containers under a category (categories nest, so “Packaging → Plastic Containers” is two linked records). Her SKU is unique **per supplier**, not globally.

A product cannot go live until three things exist:

- An **MOQ** — this is wholesale, not retail.
- Exactly one inventory record.
- At least one price tier.

Price tiers are the wholesale part: 100–499 at \$5.00, 500–999 at \$4.50, 1000+ at \$4.00. Tiers on one product may not overlap.

Current state and history are separate — and this pattern repeats throughout the system. Inventory holds only the current truth (available and reserved). The stock ledger holds every change, append-only, each row recording the movement and the before/after values. “How much do I have?” is one fast read; “why is it that number?” is a replayable log.

Note what is *not* on the product: **tax**. VAT lives in settings and is copied onto documents later. And the catalog price is only the live shop window — once a deal is agreed, the price is copied elsewhere and the catalog can change freely.

ACT 5 Ahmad wants to buy — three ways to start

Route	What happens
Direct	Ahmad browses the catalog, finds Cedar Plastics and raises a request himself.
AI Sourcing	He types “I need packaging for a small olive oil bottling operation.” A sourcing request is created with structured items, and recommendations point at <i>real</i> products and <i>real</i> supplier companies. He converts it into an RFQ.
Business Planner	He is earlier still — planning a business. A plan produces items tagged ESSENTIAL , RECOMMENDED or OPTIONAL with price estimates that record where each number came from. The plan converts to a sourcing request, then to an RFQ.

All three roads lead to the same place: the beginning of the commercial spine.

ACT 6 RFQ, quotes and haggling

Ahmad's **RFQ** says what he wants and who may supply it: a destination, a required-by date, and optionally a list of invited suppliers. Each distinct thing he wants is its own line with quantity, unit and requirements. A line may point at a catalog product — or describe something nobody has listed yet.

Cedar Plastics and two other verified suppliers each submit a **quotation** against that one RFQ. Each quotation line must point back to the specific RFQ line it answers, so that when a supplier quotes \$5 on one line and \$8 on another, there is no ambiguity about which requested product got which price. Comparing three suppliers would otherwise be guesswork.

Two separate mechanisms for haggling

Negotiation is the commercial back-and-forth: each offer references its parent offer, so Ahmad's counter points at Lara's original and hers points at his. The whole history reads as a chain. The quotation separately keeps its own version snapshots.

Conversations are the chat — and the key decision is that chat is a **general capability, not a negotiation feature**. A thread can hang off an RFQ, a quotation, a negotiation, an order, a dispute, a product enquiry, or nothing at all. A negotiation may have no chat, and a chat may have no negotiation.

Several people from each side sit in one thread, and read position, mute and leave state are **per person** — which is exactly why participants are their own records rather than a list on the conversation.

A message is never a commercial record. If someone types “ok, \$4.20 final”, that is not a price. Prices exist only on quotation lines and negotiation offers. A message may *reference* a document; it never restates its terms.

ACT 7 Accepting the quote creates the Purchase Order

Ahmad accepts Cedar Plastics' quote at \$1,000. The order that is created **is** the purchase order — there is no second order record. Shipments, invoices, payments, commission, disputes and reviews all hang off it.

The order lines are copies, and that copying is the point. Each line snapshots the product name, SKU, agreed unit price, discount and tax. If Lara renames the product tomorrow, raises her price to \$6, or deletes the tier entirely, this order still says \$5.00 and still shows the old name. The catalog answers “what does it cost today?”; the order answers “what did we agree?” They are allowed to disagree.

The moment the order is confirmed, three things fire together

Effect	Detail
Stock is reserved	In a single atomic transaction, a reservation row is written and quantity moves from available to reserved. A <code>pending</code> order reserves nothing — only confirmation does. This is what stops two buyers from both holding the last 500 units while neither has committed.
Commission is snapshotted	A pending commission record copies the rate and base from platform settings <i>at that moment</i> . Raising the fee from 5% to 6% next month leaves this order untouched.
Minimum order value is checked	Against the configured floor.

ACT 8 Shipping, then what the buyer owes

Shipments hang off the order, and one order may have several — partial deliveries are normal in wholesale. Each carries carrier, tracking number, tracking events and delivery evidence. TradeBay is not the carrier; a shipment records fulfilment, it is not a second order.

The **invoice** is issued from the confirmed order. At issue it copies the active VAT rate and name, and copies the order lines into itself. If VAT changes next year, this invoice must not change — that is the entire reason for the snapshot.

An issued invoice is never edited. Not the total, not a line, not the tax. If something is wrong, issue a **credit note** to reduce what is owed; if cash must go back, create a **refund**. The original stays visible with the correction attached.

Ahmad pays. A payment may be **allocated across several invoices** — \$6,000 might settle \$4,000 of one and \$2,000 of another. Payments and invoices are not one-to-one. When a payment completes it carries its own receipt number and document; a receipt is a view of a payment, not a separate thing.

The AR ledger

Invoices, payments, credit notes and refunds are the **source documents**. The AR ledger is the **append-only record of their effect**, written in the same transaction as the document that caused it, so a document and its financial effect can never exist separately.

Event	Direction	Effect on what the buyer owes
Invoice issued	debit	increases
Payment received — one row per allocation	credit	decreases
Credit note applied	credit	decreases
Refund processed	debit	increases
Reversal	opposite of the original	correction

There is no stored AR balance anywhere. No `amount_outstanding`, no `buyer_balance`. What the buyer owes is computed by summing posted ledger rows, and aging comes from the issue and due dates combined with that sum. A stored balance is a number that can silently disagree with the documents behind it — and once it does, nobody can tell which one is lying.

ACT 9 Where the money actually went

Customer finance asks *what does the buyer owe?* Platform money asks a different question: *when cash moved through TradeBay, where did it go?* The two are deliberately never merged. Following the \$1,000:

Step	Routing ledger row	Funds state
Buyer pays \$1,000	buyer payment +\$1,000	<code>held</code>
Order fulfilled	funds released	<code>released</code>

Step	Routing ledger row	Funds state
Fee recognized	platform fee +\$50	released
Supplier owed	payable created: \$1,000 - \$50 = \$950	—
Supplier paid	supplier payout -\$950	released

`funds_state` is internal bookkeeping, not legal escrow. The payment provider holds the regulated funds. This database keeps the marketplace audit trail of what happened. No field may be named `escrow_balance`.

The split between **payable** and **payout** is not bureaucracy — it is the difference between “we owe you \$950” and “we sent you \$950”, and those are genuinely different facts for days at a time. A payable is settled only when its payout completes. Settlement batches are an administrative convenience: group many payouts, have an admin approve the batch, pay them all.

Two numbers fall out for free: **GMV** is the sum of order totals, and **TradeBay revenue** is the sum of recognized commission. They are never the same number and are always reported separately.

ACT 10 Trust, and the end of the story

The order completes. Ahmad leaves a review — **one per order**, which is what gives ratings meaning: you cannot review a supplier you never bought from, and you cannot review the same purchase twice. The result rolls up into the supplier’s rating summary.

Throughout, notifications have been landing in people’s inboxes: RFQ published, quotation submitted, quotation accepted, order confirmed, order shipped, invoice issued, payment completed, dispute opened, dispute resolved, payout completed. A notification is the platform poking you; a message is a human talking to a human. They are not the same thing.

07 Exception Scenarios

Most designs document only the happy path. These are the branches the system must handle, and exactly what each one must do.

Scenario	Required behaviour
Nobody quotes the RFQ	It expires when the required-by date passes. Nothing happened, nothing to undo.
Buyer rejects every quote	Quotations are rejected and the RFQ may be cancelled. No order, no reservation, no money.
A quote goes stale	Past its validity date the quotation expires and cannot be accepted.
Supplier withdraws before acceptance	The quotation is withdrawn.
Order cancelled after confirmation	The reservation is released exactly once, quantity returns from reserved to available, and the cancellation is recorded in the order's status history. Any money already moved is corrected by the rows below.
Wrong amount invoiced	Do not edit the invoice. Issue a credit note; a credit posts to the AR ledger and what the buyer owes goes down. Crediting without refunding is valid — the reduced balance simply carries to the next invoice.
Buyer needs cash back	A refund against the original payment. On processing it posts a debit on the AR ledger and a refunded row on the routing ledger. The original payment is never deleted or reduced.
A ledger row was posted wrong	Insert a reversal row with the opposite direction, referencing the original. Both rows stay forever.
Payment gateway fires the same webhook twice	Unique idempotency and provider-event keys reject the duplicate. One payment, one ledger post — never three.
A payout to the supplier fails	The payout is marked failed with a reason and the payable stays open . A new payout may be attempted against the same payable; the relationship is one-to-many precisely so retries are possible.
Goods arrive damaged	A dispute is opened against the order, which may move the order to disputed. Evidence is attached. A Platform Operator resolves it, usually triggering a credit note or refund, and the related payable may be held out of settlement while contested.
Two buyers race for the last 500 units	Whichever order confirms first wins, because reservation and stock update happen in a single atomic transaction. The second is rejected. Pending orders never reserve, so nobody can hold stock hostage without committing.
An employee leaves	Their membership is removed or suspended. Their user account survives, along with memberships at other companies and every document they created. History is never orphaned.

Scenario	Required behaviour
Buyer overpays	The excess is flagged and left unallocated. It must never silently reduce an unrelated invoice.
VAT rate changes	Already-issued invoices keep their snapshotted rate. Only new invoices use the new rate.
Commission rate changes	Already-confirmed orders keep their snapshotted rate and base.

08 Functional Requirements

All requirements below are **Must** unless marked otherwise.

8.1 Identity, membership & authorization

ID	Requirement
FR-AUTH-01	Users register with email and password.
FR-AUTH-02	Email verification is required before commercial writes.
FR-AUTH-03	Password recovery uses expiring secure tokens.
FR-AUTH-04	Authorization uses atomic <code>resource.action</code> permission codes resolved server-side on every request — never a role-name comparison.
FR-AUTH-05	Sessions support refresh rotation, revocation and expiry.
FR-AUTH-06	Admins can suspend accounts with a reason and an audit record.
FR-AUTH-07	Security-sensitive actions record actor, action, entity, timestamp, IP and metadata.
FR-AUTH-08	One person may belong to several businesses with a different role in each, and switches the active business context.
FR-AUTH-09	Invitations carry a role , never individual permissions. Accepting creates a membership.
FR-AUTH-10	A user may only grant a permission set that is a subset of their own.
FR-AUTH-11	Five system roles are seeded per business and cannot be deleted. Custom roles may be added.
FR-AUTH-12	Platform staff operate through the platform business account and platform-wide permissions, not a special user flag.

8.2 Supplier onboarding & verification

ID	Requirement
FR-BIZ-01	Businesses maintain legal and display name, type, tax number, contact and address.
FR-BIZ-02	A company may buy immediately; selling requires a supplier profile — exactly one per company.
FR-BIZ-03	Suppliers submit verification documents and declare service areas.
FR-BIZ-04	Platform staff approve, reject or suspend verification with a reason.
FR-BIZ-05	Every selling write requires the permission code and a verified profile.
FR-BIZ-06	Only verified suppliers display the verified badge.

ID	Requirement
FR-BIZ-07	The profile carries a rating summary recomputed from published reviews.

8.3 Catalog, pricing & inventory

ID	Requirement
FR-PROD-01	Products move through draft, active and inactive.
FR-PROD-02	SKU is unique per supplier , not globally.
FR-PROD-03	Quantity-based price tiers are non-overlapping.
FR-PROD-04	A product cannot go active without an MOQ, an inventory record and at least one price tier.
FR-PROD-05	Available and reserved quantities are updated atomically to prevent overselling.
FR-PROD-06	Every stock movement writes an append-only ledger row with type, quantity, cause and before/after values.
FR-PROD-07	Quantities are validated against MOQ and current pricing rules.
FR-PROD-08	Categories nest, with activation state and display order.
FR-PROD-09	Products support multiple ordered images.
FR-PROD-10	Tax is never a product attribute.

8.4 Discovery & comparison

ID	Requirement
FR-DISC-01	Buyers search products and suppliers by text.
FR-DISC-02	Filters cover category, price, MOQ, supplier location, verification and availability.
FR-DISC-03	Sorting by relevance, price, rating and lead time.
FR-DISC-04	Buyers compare competing offers side by side on objective commercial attributes.

8.5 RFQ, quotation & negotiation

ID	Requirement
FR-RFQ-01	Buyers create an RFQ with title, description, destination, required-by date and line items.
FR-RFQ-02	An RFQ line may reference a catalog product or describe something unlisted.
FR-RFQ-03	Buyers invite named suppliers or publish openly, tracking response status.

ID	Requirement
FR-RFQ-04	Suppliers submit itemized price, quantity, lead time, discount, charges, tax, validity and terms.
FR-RFQ-05	Every quotation line references the specific RFQ line it answers, unique per quotation, so prices cannot be misattributed.
FR-RFQ-06	One RFQ collects many competing quotations.
FR-RFQ-07	Every commercial revision is retained; exactly one accepted version creates the order.
FR-RFQ-08	RFQs and quotations expire on configured dates.
FR-NEG-01	Buyer and supplier exchange structured offers carrying quantity, price, delivery and payment terms.
FR-NEG-02	Each counter-offer references its parent, so the full history is reconstructable.
FR-NEG-03	A negotiation anchors on the RFQ or quotation, not on a conversation. Attaching a thread is optional in both directions.
FR-NEG-04	An accepted offer updates the quotation and appends a commercial version.
FR-NEG-05	No separate Deal entity. Quotation and order remain the commercial source of truth.

8.6 Orders & fulfilment

ID	Requirement
FR-ORD-01	The accepted quotation creates the order. The order is the purchase order; there is no second order record.
FR-ORD-02	Order lines snapshot product name, SKU, unit price, discount and tax at agreement. Later catalog edits cannot change them.
FR-ORD-03	Confirmation atomically reserves stock and creates a pending commission record. A pending order reserves nothing.
FR-ORD-04	Order status history records every transition with actor and timestamp.
FR-ORD-05	Shipments carry carrier, tracking reference, tracking events and delivery evidence. One order may have several.
FR-ORD-06	Cancellation releases the reservation exactly once and triggers any required financial correction.
FR-ORD-07	One currency per order.
FR-ORD-08	Orders are validated against the configured minimum order value.

8.7 Communication

ID	Requirement
FR-MSG-01	A thread runs between two companies, optionally about an RFQ, quotation, negotiation, order, dispute or product — or about nothing at all.
FR-MSG-02	Several users per side; read position, mute and leave state are per user .
FR-MSG-03	Messages may be text, attachment, reference or system, with optional reply threading.
FR-MSG-04	A message is never a commercial record. A reference message deep-links the authoritative document and nothing more.
FR-MSG-05	Access requires an active participant record and active membership in one of the two companies.
FR-MSG-06	Unread counts are computed from the last-read timestamp. No stored counter.
FR-MSG-07	One thread per commercial context. Direct and support threads are exempt.
FR-MSG-08	Messages are never hard-deleted; editing and removal are marked. Threads are dispute evidence.
FR-MSG-09	Platform events are written into the thread as system messages so it reads as one timeline.
FR-MSG-10	Support is a thread whose counterparty is the platform company. No separate ticket system.

8.8 Trust & inbox

ID	Requirement
FR-TRUST-01	Only a completed order may be reviewed, and only once.
FR-TRUST-02	Reviews carry a rating and comment, and may be moderated or removed with an audit trail.
FR-TRUST-03	Either party may open a dispute on an order, attaching evidence.
FR-TRUST-04	Platform staff resolve disputes with a recorded resolution and notes.
FR-TRUST-05	Notifications are in-app, scoped to user and business, and deep-link the originating resource.
FR-TRUST-06	Notified events include: RFQ published, quotation submitted, quotation accepted, order confirmed, order shipped, invoice issued, payment completed, dispute opened, dispute resolved, payout completed.

8.9 Administration

ID	Requirement
FR-ADM-01	Platform staff manage users, businesses, suppliers, products and categories.
FR-ADM-02	Platform staff moderate reviews and resolve disputes.
FR-ADM-03	Authorized finance users view invoices, payments, receivables, payables, payouts and settlement batches.
FR-ADM-04	Platform staff configure commission, VAT and business letterhead.
FR-ADM-05	Dashboards show GMV and recognized commission revenue as separate figures.
FR-ADM-06	High-impact administrative actions are immutable audit events.

8.10 System settings

ID	Requirement
FR-CFG-01	Platform settings hold commission rate and base, default currency, minimum order value and the payment provider name .
FR-CFG-02	VAT is configurable with one active rate and effective dates. The Lebanese rate is never hardcoded.
FR-CFG-03	Business settings hold the letterhead: name, contact, tax registration number, invoice prefix.
FR-CFG-04	Settings hold what is true <i>now</i> ; documents keep copies of what was true when created. Changing a setting must never rewrite an issued invoice or a recognized fee.
FR-CFG-05	Provider API keys live in environment or secret management, never in the database.

The snapshot principle. Live value → frozen copy: VAT rate → invoice tax rate · commission rate → commission record · catalog price → order line unit price · product name and SKU → order line snapshots. If you can change history by editing a settings page, you do not have an auditable system.

09 Finance Requirements

9.1 Customer accounts receivable

ID	Requirement
FIN-AR-01	A confirmed order produces a customer invoice.
FIN-AR-02	The invoice stores subtotal, discounts, charges, tax and total with deterministic rounding, and snapshots the active VAT rate and name at issue.
FIN-AR-03	Invoices support due dates and configured payment terms.
FIN-AR-04	Outstanding = invoice total – valid payments – valid credits and refunds, computed from the ledger .
FIN-AR-05	Several payments may apply to one invoice, and one payment may be allocated across several invoices.
FIN-AR-06	Aging buckets: current, 1–30, 31–60, 61–90 and 90+ days.
FIN-AR-07	Overpayment is flagged. An unallocated remainder must never silently reduce an unrelated invoice.
FIN-AR-08	Buyer statements show invoices, payments, credits and a running balance.

9.2 Payments & corrections

ID	Requirement
FIN-PAY-01	Every payment is a separate record with a unique reference.
FIN-PAY-02	Payment states: pending, then completed, failed or reversed.
FIN-PAY-03	Provider callbacks are idempotent. The same provider event must never create a second completed payment or a second ledger post.
FIN-PAY-04	Payments allocate to eligible invoices; each allocation posts its own ledger row.
FIN-PAY-05	A completed payment carries a receipt number and document. There is no separate receipt record.
FIN-COR-01	Issued invoices are never edited. Corrections use credit notes.
FIN-COR-02	A credit note may reduce what is owed without any refund.
FIN-COR-03	A refund references the original payment and never deletes or reduces it.

9.3 The AR ledger

ID	Requirement
FIN-LED-01	Every material financial movement posts a ledger row with source, amount, currency, direction, date and reference, written in the same database transaction as its source document.
FIN-LED-02	Posted rows are never updated or deleted. Corrections insert a reversal row referencing the original.
FIN-LED-03	Every row links back to an invoice, payment, credit note or refund.

9.4 Commission, payables & payouts

ID	Requirement
FIN-REV-01	Administrators configure the commission rate and base.
FIN-REV-02	Rate and base are copied onto the commission record when the order is confirmed . Later rate changes must not rewrite historical fees.
FIN-REV-03	Commission becomes revenue when held funds are released, not when the order is created.
FIN-REV-04	GMV is the sum of order totals; revenue is recognized commission. They are reported separately.
FIN-REV-05	Refunds and reversals create adjusting records rather than deleting the original.
FIN-AP-01	Net payable = gross – commission ± adjustments. The payable is the accounts-payable document; there is no separate supplier bill.
FIN-AP-02	Payable states: open, partially settled, settled, disputed.
FIN-AP-03	Supplier statements show gross sales, commission, adjustments, settlements and outstanding balance.
FIN-SET-01	A payout is the actual transfer. A payable is settled only when its payout completes.
FIN-SET-02	A failed payout records the reason and leaves the payable open so a retry can be attempted.
FIN-SET-03	Finance may group payouts into an approval batch.
FIN-SET-04	Batches require explicit approval before payment.
FIN-SET-05	Disputed payables are held out of settlement.
FIN-SET-06	The same provider event must never pay out twice.

9.5 The routing ledger

ID	Requirement
FIN-PLT-01	Every movement of money through the platform posts an append-only row: buyer payment, funds released, platform fee, supplier payout, refund, adjustment.
FIN-PLT-02	<code>held → released / refunded</code> is TradeBay's internal money lifecycle. It is explicitly not legal escrow ; the provider holds regulated funds.
FIN-PLT-03	Held, released, paid and refunded positions are summed from posted rows. No writable platform or supplier balance exists.
FIN-PLT-04	Posted rows are reversed with a new row, never edited.

9.6 Worked example

Event	Amount	Business effect
Customer invoice issued	\$1,000	Buyer owes \$1,000 — AR ledger debit.
Buyer payment completed	\$1,000	AR cleared (credit). Routing ledger: <code>held</code> .
Order fulfilled	—	Funds <code>released</code> .
Commission recognized	\$50	TradeBay revenue. 5% of the configured base.
Supplier payable created	\$950	TradeBay owes the supplier.
Supplier payout completed	\$950	Payable settled. Routing ledger $-\$950$.

GMV \$1,000. TradeBay revenue \$50. Reporting the first as the second is the single most common financial error in marketplace software, and the model exists to make it impossible.

10 AI Capabilities

Three capabilities. All are advisory, and all inherit normal authorization and business isolation.

10.1 AI Sourcing Assistant

A buyer describes a need in natural language: *"I need 500 boxes of olive oil delivered to Beirut within 10 days."*

ID	Requirement
FR-AI-01	Accept a natural-language sourcing request.
FR-AI-02	Convert it into editable structured requirements: product, quantity, unit, destination, timing.
FR-AI-03	Recommendations must reference real products and real supplier companies from the buyer's authorized scope. The model turns language into structure; the marketplace search supplies the matches.
FR-AI-04	A completed sourcing request converts into an RFQ with its line items.
FR-AI-05	<i>[Should]</i> Each recommendation carries a plain reason — category match, MOQ compatibility, prior purchase.
FR-AI-06	Recommendations are persisted, so re-rendering never re-invokes the provider.

10.2 Personalized Product Recommendations

ID	Requirement
FR-AI-07	Recommend active products from permitted buyer signals: previous purchases, engaged products, frequently used categories.
FR-AI-08	Price, MOQ, availability and verification always come from platform records, never from the model.
FR-AI-09	Users may ignore recommendations without losing normal catalog access.

10.3 Business Planner

ID	Requirement
FR-AI-10	From a described business type, generate plan items classified ESSENTIAL , RECOMMENDED or OPTIONAL .
FR-AI-11	Price estimates derive from real marketplace prices where matching products exist, and each estimate records its source as marketplace, historical or AI estimate.
FR-AI-12	A saved plan converts into a sourcing request, and from there into an RFQ.

10.4 Decision boundary

AI may	AI may not
Interpret natural-language sourcing needs	Approve or verify a supplier
Draft RFQ wording and structure	Bypass supplier verification
Recommend products and suppliers	Override authoritative price, stock or order state
Personalize discovery	Award a quotation or create a purchase order
Summarize known trade-offs	Create or change financial truth
Explain why something was shown	Bypass roles, permissions or business isolation

FR-AI-13	AI must never present a generated assumption as a verified fact, and must never complete a commercial action without explicit user confirmation.
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11 Core Business Rules

ID	Rule
BR-01	Only verified suppliers may publish products or sell.
BR-02	Orders cannot fall below product MOQ without an authorized exception.
BR-03	The applicable price tier is chosen from the final confirmed quantity and preserved on the order.
BR-04	Pending orders do not consume sellable stock. Confirmed orders reserve stock atomically.
BR-05	Eligible cancellation releases the reservation exactly once.
BR-06	Only one accepted quotation version may generate an order.
BR-07	Later catalog edits cannot change a historical order price, name or quantity.
BR-08	Issued invoices are never silently edited. Corrections use credit notes.
BR-09	Payment status derives from payment records and provider events, never a manually toggled flag.
BR-10	Refunds reference the original payment and never erase it.
BR-11	Commission is calculated from the snapshotted rule and is separate from supplier gross sales.
BR-12	Supplier settlement equals the eligible amount after commission and approved adjustments.
BR-13	GMV is not profit. TradeBay revenue is recognized commission.
BR-14	Customer and supplier balances are calculated from documents and posted transactions, never stored as mutable totals.
BR-15	High-impact actions and financial state changes are audit logged.
BR-16	AI cannot create unsupported marketplace facts and requires confirmation before commercial actions.
BR-17	All workflows are designed for domestic Lebanese B2B trade.
BR-18	Only eligible completed transactions may create reviews, one per order.
BR-19	A message is never a commercial record.
BR-20	A user acts only within the scope of their role and active business.
BR-21	Posted rows in either ledger are corrected by reversal, never by edit or delete.
BR-22	A payable is settled only when its payout completes; a failed payout leaves it open.

The eight laws, condensed. 1. People log in, companies trade, memberships connect them. 2. Permissions are atomic codes on roles, resolved fresh on every request. 3. Buying is open; selling requires verification. 4. Current state and history are separate. 5. Agreed commercials are frozen. 6. Financial documents are never edited or deleted. 7. Balances are never stored, always computed. 8. Two ledgers, never merged — one answers what the buyer owes, the other where the cash went.

12 State Lifecycles

Domain	States
User	pending → active → suspended / deactivated
Membership	invited → active → suspended / removed
Invitation	pending → accepted / expired / revoked
Supplier verification	unverified → pending → verified / rejected
Product	draft → active → inactive
RFQ	draft → published → responding → negotiating → awarded / cancelled / expired
Quotation	draft → submitted → negotiating → accepted / rejected / expired / withdrawn
Order	pending → confirmed → processing → shipped → delivered → completed , plus cancelled / disputed
Shipment	pending → preparing → shipped → in_transit → out_for_delivery → delivered / failed
Invoice	draft → issued → partially_paid → paid / overdue / void / credited
Payment	pending → completed / failed / reversed
Credit note	draft → issued → applied / cancelled
Refund	requested → approved → processed / rejected
AR ledger row	posted / reversed — append-only
Commission	pending → recognized / reversed
Supplier payable	open → partially_settled → settled / disputed
Supplier payout	pending → processing → completed / failed
Settlement batch	draft → pending_approval → approved → paid / failed / cancelled
Routing ledger row	posted / reversed ; funds state held → released / refunded
Dispute	open → under_review → resolved / closed

Domain	States
Review	draft → published / moderated / removed
Negotiation	OPEN → IN_PROGRESS → AGREED / REJECTED / EXPIRED / CANCELLED
Negotiation offer	PROPOSED → COUNTERED → ACCEPTED / REJECTED / EXPIRED
Conversation	ACTIVE → CLOSED / ARCHIVED
Sourcing request	DRAFT → SEARCHING → COMPLETED → CONVERTED_TO_RFQ
Business plan	DRAFT → GENERATED → SAVED → CONVERTED_TO_SOURCING

13 Non-Functional Requirements

Category	Requirement
Security	Adaptive password hashing; HTTPS in production; httpOnly secure cookies; refresh tokens hashed at rest; server-side authorization on every protected operation; input validation; rate limiting; uploads type/size validated and stored outside executable paths.
Data integrity	Financial totals, inventory reservations and state transitions are deterministic and protected against duplicate processing. Money uses exact decimal types — never binary floats.
Atomicity	Operations that must not partially apply — stock reservation, ledger posting — run inside a database transaction together with their source document.
Auditability	High-impact operational changes and all posted financial changes trace to actor, source and time.
Performance	Common marketplace pages respond under expected load; large collections are paginated and indexed.
Scalability	Domain boundaries are separated so marketplace, finance and messaging evolve independently.
Observability	Errors, domain events and financial failures are logged with correlation identifiers and without secrets.
Maintainability	Modular architecture, validation schemas, service-layer business logic, automated tests on money and inventory rules.
Accessibility	Forms, tables and navigation are keyboard usable with semantic structure.
Localization	Currency, dates and addresses are configurable rather than hardcoded.

14 Out of Scope

Explicitly **not** built. Several appeared in BRD v1.1 and are withdrawn.

Excluded	Reason
Chart of accounts, journal entries, accounting periods (<code>FIN-LED-04..06</code>)	Statutory accounting. TradeBay keeps two operational ledgers, not a general ledger.
Platform operating expenses and P&L	Internal operations, not marketplace commerce.
Loyalty points, tiers, rewards, badges	Not wholesale procurement.

Excluded	Reason
Favorites	Nice-to-have.
Separate supplier bill records	The payable is the AP document.
Separate receipt records	Receipt fields live on the payment.
Separate verification case records	Status and documents live on the supplier profile.
Separate rating records	Rating and comment live on the review.
Commission rule records	A configured rate plus a snapshot on each commission record.
Tax engine, jurisdictions, exemptions, filings	One configurable VAT rate plus an invoice snapshot.
Notification templates and delivery logs	Templates in code; in-app notifications only in v1.
Bank accounts, treasury, FX, reconciliation, legal escrow	TradeBay is not a bank. The provider moves regulated funds.
Warehouses, procurement budgets, requisitions, vendor contracts	Not in the procurement flow.
Per-user permission overrides	Grants resolve through roles.
Real-time message transport	The model is complete; only the transport is deferred. Polling is acceptable for v1.
Group threads across three or more companies, reactions, presence, typing indicators	Conversations are two-company by design.
A separate Deal entity	Quotation and order are the commercial truth.
Freight, customs and carrier operations	TradeBay is not a logistics provider.

15 MVP Acceptance Criteria

- A buyer registers, verifies email, creates a business and accesses buyer functions.
- A supplier submits verification and **cannot sell** until a platform operator approves it.
- A Business Admin invites an employee by role, and that employee's permissions resolve from the role on every request.
- A supplier publishes a product with MOQ, inventory and non-overlapping price tiers.
- A buyer uses the AI Sourcing Assistant, receives recommendations that reference **real** catalog products, and converts them into an RFQ.
- Two suppliers quote the same RFQ and the buyer compares them side by side.
- The parties negotiate through counter-offers and discuss in a thread.
- Accepting a quotation creates **one** purchase order with frozen commercial snapshots.
- Confirmation reserves stock exactly once and cannot oversell under concurrent orders.
- The invoice carries a VAT snapshot and posts a debit to the AR ledger.
- A payment allocates across invoices and posts one credit per allocation.
- **Outstanding balance and buyer statement are computed from ledger rows, with no stored balance field anywhere in the system.**
- Commission is recognized on release, a payable is created for the net, and a completed payout settles it; a failed payout leaves it open.
- Held, released and paid positions reconcile by summing the routing ledger.
- Refunds and credit notes correct the record without deleting the original.
- A buyer reviews a completed order, once.
- Duplicate provider events do not double-charge, double-post or double-pay.
- AI cannot invent price, MOQ, inventory, verification, order or financial facts, and cannot complete a commercial action without explicit confirmation.

16 Glossary

Term	Meaning
MOQ	Minimum order quantity.
RFQ	Request for quotation — a buyer's request for commercial pricing and terms.
Quotation	A supplier's offer against an RFQ.
Purchase order	The binding agreement created from an accepted quotation.
Snapshot	A copy of a value taken at the moment of agreement, immune to later configuration changes.
GMV	Gross merchandise value — total marketplace transaction value. Not TradeBay revenue.
AR	Accounts receivable — what buyers owe TradeBay.

Term	Meaning
AP	Accounts payable — what TradeBay owes suppliers.
Commission	TradeBay's fee on an eligible order, recognized when funds are released.
Payable	The amount TradeBay owes a supplier.
Payout	The actual transfer of that amount.
Settlement batch	An administrative grouping of payouts for one approval.
Funds state	TradeBay's internal money lifecycle — held, released, refunded. Not legal escrow.
Ledger	An append-only record of financial movements. TradeBay has two: AR and routing.
Reversal	A new ledger row that undoes a previous one. The original always remains visible.

END OF BUSINESS REQUIREMENTS DOCUMENT

TRADEBAY · SOURCE · CONNECT · GROW